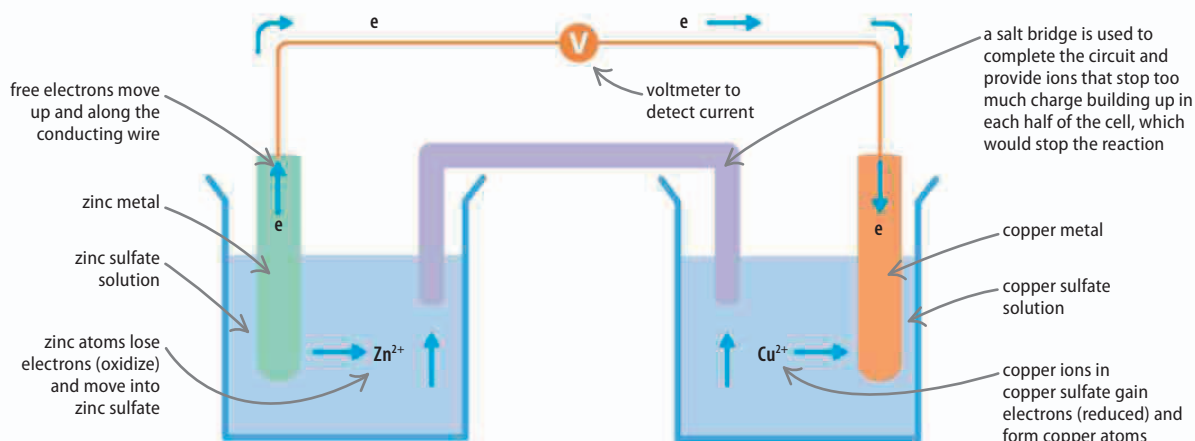
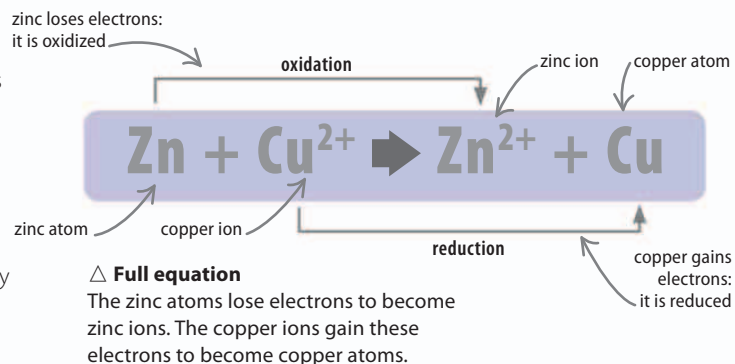


Electrochemistry

The exchange of electrons that occurs in redox reactions can be used to create an electric current in an apparatus called an electrochemical cell. The current forms when electrons are released from the oxidation reaction and made to travel to the reduction reaction, which will absorb the electrons. In this experiment, a piece of zinc metal is dipped in zinc sulfate and a piece of copper is put in copper sulfate. These two metals are connected by a conducting wire. The sulfate solutions have free ions, which can carry an electric current (see page 148).



△ Oxidation cell

In the left half of the cell, oxidation occurs. The zinc metal atoms lose electrons to form zinc ions. The free electrons travel up and along the conducting wire to the reduction cell. The zinc ions (Zn^{2+}) move into the zinc sulfate solution.

△ Reduction cell

In the right half of the cell, reduction happens. Copper ions (Cu^{2+}) in the copper sulfate solution move to the piece of copper and accept two electrons each that have arrived from the oxidation cell. The copper ions thus become copper atoms.

Corrosion

A familiar phenomenon involving redox reactions is corrosion, in which metals and other materials are oxidized. Corrosion takes place in damp conditions, involving a reaction with oxygen or carbon dioxide and occasionally with pollutants such as hydrogen sulfide.

▷ Types of corrosion

The products of the reaction, such as rust, cause discolouration and weaken the original object.

Metal	Corrosion	Chemical name	Description
iron	rust	hydrated iron oxide	flaky rust expands and cracks the metal
copper	verdigris	copper carbonate	turns objects gray-green
aluminum	alumina	aluminum oxide	forms a dull layer on metal
silver	tarnish	silver sulfide	makes silver dark and dull
gold	no corrosion	none	gold always stays shiny